

THE GENERALISED PARTIAL CORRELATION MATRIX

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Let X be a p -dimensional random vector that has a multivariate normal distribution with mean μ and covariance matrix Σ . Suppose X is partitioned into three subvectors $(X^{(1)'} X^{(2)'} X^{(3)'})'$ with q , r and s components respectively, where $q+r+s = p$. Let $\mu(p \times 1)$ and $\Sigma(p \times p)$ be partitioned accordingly. Suppose that the third set of components $X^{(3)}$ is fixed at some value $x^{(3)}$. From the properties of the multivariate normal distribution it follows that the conditional distribution of $(X^{(1)'} X^{(2)'})' | x^{(3)'}$ will be a $(q+r)$ multivariate normal distribution with covariance matrix $\Sigma_3 = \begin{bmatrix} \Sigma_{11.3} & \Sigma_{12.3} \\ \Sigma_{22.3} & \Sigma_{22.3} \end{bmatrix}$

where $\Sigma_{11.3}$ is $q \times q$, $\Sigma_{12.3}$ is $q \times r$ and $\Sigma_{22.3}$ is $r \times r$ (Anderson 1958 page 33). Measures of the relationships that exist between $X^{(1)}$ and $X^{(2)}$ when $X^{(3)}$ is fixed at some value $x^{(3)}$ are of considerable theoretical and practical interest. The most familiar measure of partial association is the partial correlation coefficient which measures the relationship between two components of a random vector when the effects of some or all the others have been removed. Other measures which have been defined are

- (i) The partial multiple correlation coefficient (Roy and Whittlesey 1952).
- (ii) The partial trace correlation coefficient (Hooper 1962).
- (iii) Canonical partial correlation coefficients (Roy and Whittlesey 1952).

Khatri (1964) defined the generalised multiple correlation matrix P and its estimate R and its properties were discussed by Troskie (1969). All ordinary correlation coefficients can be derived from this matrix by considering special cases. By direct analogy, a similar matrix, which will embody all types of partial correlation coefficients, can be defined using the submatrices of Σ_3 . Accordingly, we define the generalised partial correlation matrix as

$$P_3 = \Sigma_{11.3}^{-1/2} \Sigma_{12.3} \Sigma_{22.3}^{-1} \Sigma_{21.3} \Sigma_{11.3}^{-1/2}$$

The following results can be written down immediately from the corresponding results of Troskie (1969). For $q = 1$, $r = 1$, P_3 becomes the square of the partial correlation coefficient. For $q = 1$, $r > 1$, P_3 is the square of the partial multiple correlation coefficient. The non-zero roots of P_3 are the squares of the partial canonical correlations and if P_3 has q non-zero roots, the square of the partial trace correlation coefficient is $\frac{1}{q} \text{tr } P_3$. These measures of partial association do not depend on the underlying distribution of the variables, but except in the case of a multivariate normal distribution, it would appear that the sampling distribution of their estimates present intricate problems.

Let $X(a)$ $a = 1, 2, \dots, N$, $N > p$ be a random sample from a $N_p(\mu, \Sigma)$ population, where each $X(a)$ is partitioned as the vector X . Let A be the Wishart matrix of this sample and

suppose A is partitioned according to the partitioning of Σ . Then by the invariance property of maximum likelihood estimates, the maximum likelihood estimate of Σ_3 is given by $\hat{\Sigma}_3 = \frac{1}{N} A_3$ where $A_3 = \begin{vmatrix} A_{11.3} & A_{12.3} \\ A_{21.3} & A_{22.3} \end{vmatrix}$ and $A_{ij.3} = A_{ij} - A_{i3} A_{33}^{-1} A_{3j}$, $i, j = 1, 2$.

The maximum likelihood estimate of P_3 is given by

$$R_3 = A_{11.3}^{-\frac{1}{2}} A_{12.3}^{-1} A_{22.3}^{-\frac{1}{2}} A_{21.3} A_{11.3}$$

and R_3 has the same properties in the sample as P_3 in the population. Under the hypothesis $P_3 = 0$ the sampling distribution of R_3 can be easily found. By Anderson (1958) Theorem 4.3.2 A_3 is distributed as $W(\Sigma_3, n-s)$ where $n = N-1$ and s is the number of variables held fixed. Using this fact the distribution of R_3 can be written down immediately from the corresponding results of Troskie (1966, 1969). Thus R_3 has a multivariate Beta Type I distribution with parameters $\frac{1}{2}r$ and $\frac{1}{2}(n-s-r)$. The density function

$$f(R_3) = \frac{\Gamma_q(\frac{1}{2}(n-s))}{\Gamma_q(\frac{1}{2}r)\Gamma_q(\frac{1}{2}(n-s-r))} |R_3|^{\frac{1}{2}(r-q-1)} |I-R_3|^{\frac{1}{2}(n-s-r-q-1)}$$

$$R_3 > 0, (I-R_3) > 0$$

where

$$\Gamma_q(\frac{1}{2}n) = \pi^{\frac{1}{2}q} q(q-1) \prod_{i=1}^q \Gamma(\frac{1}{2}(n-i+1)).$$

An easy integration yields the moments of R_3 which are given by

$$E|R_3|^h = \frac{\Gamma_q(\frac{1}{2}(n-s))\Gamma_q(\frac{1}{2}(r+2h))}{\Gamma_q(\frac{1}{2}r)\Gamma_q(\frac{1}{2}(n-s-2h))}$$

Substituting $q=1$, $r=1$ and $s \geq 1$ in the density function of R_3 we obtain the density function of the square of the partial correlation coefficient. For $q=1$, $r \geq 1$ and $s \geq 1$ we obtain the density function of the square of the partial multiple correlation coefficient. From this result it is easily seen that the distribution of the partial multiple correlation coefficient, based on a sample of size N from a $N_p(\mu, \Sigma)$ population where s of the variables are fixed, is the same as the distribution of the multiple correlation coefficient based on a sample of size $N-s$ from a $(p-s)$ dimensional normal population.

R_3 can be used to test if two sets of variables are conditionally independent. If two sets of variables are conditionally independent $\Sigma_{12.3} = 0$ and all the partial canonical correlations are zero. Four tests of the hypotheses of independence have been proposed in the literature, all of which are based on some function of the canonical correlations. Since Roy (1957 page 41) has shown that the joint distribution of the squares of the

partial canonical correlations is the same as that of the squares of the canonical correlations with N (the sample size) replaced by $N-s$, it follows that the same four tests could be used to investigate conditional independence with appropriate adjustment of the parameters. Thus tests of conditional independence will be based on the characteristic roots of R_3 since these roots are the squares of partial canonical correlations.

If $r_1^2 \geq r_2^2 \geq \dots \geq r_q^2$ are the characteristic roots of R_3 any of the following four tests could be used

(1) Wilks' Likelihood Criterion

$$W^{(q)} = \prod_{i=1}^q (1-r_i^2) = |I-R_3|$$

(2) Roy's largest Root Criterion based on the largest partial canonical correlation.

(3) Pillai's Criterion

$$U^{(q)} = \sum_{i=1}^q r_i^2 / (1-r_i^2) = \text{tr } R_3 (I-R_3)^{-1}$$

(4) Pillai's Criterion

$$V^{(q)} = \sum_{i=1}^q r_i^2 = \text{tr } R_3$$

A discussion of these tests in the unconditional case and a comparison of their power when $q = 2$ is given in Pillai and Jayachandran (1967) and Troskie (1969a).

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